

TouchPro LIS2A Host Communication Protocol

(Rev. 14)



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1. Introduction

1.1 Purpose

This document defines the interface and behaviour between a Laboratory Information Management System referred to as 'LIMS' and an ELITech clinical system referred to as 'analyser'. In order to use the analyser both embedded and application software is used, embedded software is referred to as firmware and application software is referred to as PCA (PC Application).

1.2 Scope

The scope of this document is limited to describing the ELITech clinical system implementation of the CLSI LIS standard. This document should be used together with CLSI document LIS1-A (reference [1]) and CLSI document LIS2-A (reference [2]) for a complete reference. This version is applicable for TouchPro release 2.5.

1.3 Definitions, acronyms and abbreviations

Abbreviation	Description
LIMS	Laboratory Information Management System.
PCA	PC Application.
Definition	Description
Analyser	an ELITech clinical system that performs automated testing, typically receives samples and produces tests.
Transfer phase	Is the phase between a <ENQ> and <EOT> message in which data is exchanged.
Field	One specific attribute of a record which may contain aggregates of data elements further refining the basic attribute.
Record	An aggregate of fields describing one aspect of the complete message.
Message	A textual body of information consisting of a header (H) record through a message terminator (L) record.
Frame	Eventual data that is communicated between PCA and LIMS consisting of a message with data communication specific additions.
Installed reagent	Reagent that to the knowledge of the PCA is loaded upon the reagent rotor.

1.4 References

- [1] CLSI. Standard Specification for Low-Level Protocol to Transfer Messages Between Clinical Laboratory Instruments and Computer Systems. CLSI document LIS1-A [ISBN 1-56238-489-9]. CLSI, 940 West Valley Road, Suite 1400, Wayne, Pennsylvania 19087-1898 USA, 2003.
- [2] CLSI. Standard Specification for Transferring Information Between Clinical Laboratory Instruments and Computer Systems. CLSI document LIS2-A [ISBN 1-56238-490-2]. CLSI, 940 West Valley Road, Suite 1400, Wayne, Pennsylvania 19087-1898 USA, 2003.

1.5 Overview

Chapter two describes the physical layer setup between LIMS and PCA for serial and TCP/IP.
Chapter three describes the data link layer setup used between LIMS and PCA.
Chapter four describes the general message format that is used between LIMS and PCA.
Chapter five describes the messages that are send between LIMS and PCA.
Appendix A explains numerous examples of complete frames being transmitted between LIMS and PCA.
Appendix B lists the result flags to be expected to be send from the PCA to the LIMS.

2. Hardware interfaces

2.1 Serial port interface

The characteristics for the physical layer of the communication are defined in CLSI LIS1-A standard (see [1]), Section 5 Physical Layer for Serial Binary Data Exchange. The line signal definition is shown in the following table:

DB-9 pin #	DB-9 pin #		Signal name	Direction	Remarks
5	5	SG	Signal Ground		
2	3	TD	Transmit Data	Output	Data from PCA to LIMS
3	2	RD	Receive Data	Input	Data from LIMS to PCA
8	7	RTS	Request To Send	Output	*
7	8	CTS	Clear To Send	Input	*

- The only pins really used are TX and RX, the usage of the others (marked with *) depends on the settings of the COM-port in the PCA.
- The default setting for communication will be 9600 baud, 8 data bits, no parity and 1 stop bit (8-N-1).

Warning:

The XON-XOFF protocol (which can be set by Windows) must **NOT** be used, because this can interfere with the protocol.

2.2 Network interface

The characteristics for the physical layer of the communication are defined in CLSI LIS1-A standard (see [1]), Section 7 Physical Layer for TCP/IP Data Interchange. Configuration of the TCP/IP network adapter of the analyser is outside the scope of this document, for instructions on how to setup the TCP/IP network adapter refer to the analyser manual.

For LIMS communication the PCA allows setting the LIMS IP address and port number to define the LIMS socket. The PCA will connect to the LIMS socket after a reset of the analyser or as soon as the communication is enabled in system settings.

The connection will remain active as long as the PCA is running on the analyser-PC. If for some reason the TCP/IP connection is lost (e.g., because the LIMS system was restarted), then the PCA will automatically (continuously) try to reconnect.

3. Data handling

3.1 General

The properties for the message sending and receiving are defined in the CLSI LIS1-A standard (see [1]), Section 6 Data Link Layer. The PCA implements the CLSI LIS1-A standard, this section is intended to provide a short summary/introduction into the CLSI LIS1-A standard. For detailed information on error handling, time-out behaviour and connection setup/maintenance refer to the CLSI LIS1-A standard (see [1]).

The CLSI LIS1-A standard defines a line-bidding protocol to be setup between LIMS and PCA. This line-bidding protocol will identify a party as the receiver and a party as sender during the coming communication. A system ready to send information will transmit an <ENQ> character. The receiver will reply with an <ACK> if it is ready to receive or an <NAK> when it is busy. See the state diagram in Annex A1 of the CLSI LIS1-A standard (See [1]).

After the sender and the receiver have been identified, the sender will transmit frames. Each frame starts with an <STX> character and ends with an <ETB> or <ETX> character, a checksum and <CR> <LF> characters. The content of a frame is the message that is to be sent (see section 5).

The receiver will acknowledge the frame by responding with an <ACK> or <EOT>, or the receiver may indicate an error by responding with an <NAK>. Upon receiving of an <NAK> the sender retransmits the last frame. The <EOT> character indicates that the receiver has information to send and requests that the line is returned to idle state. The sender may honour the <EOT> by responding with an <EOT> or it may ignore the <EOT> and send the next frame.

After the sender has transmitted an <EOT> character, the line is returned into idle state.

3.2 Frame format

Messages are sent in frames, each frame contains a maximum of 64,000 characters (including frame overhead). Messages longer than 64,000 characters are divided between two or more frames. Multiple messages are never combined in a single frame. Every message must begin in a new frame. A frame is one of two types, an intermediate frame or an end frame. A message containing 64,000 characters or less is sent in a single end frame, longer messages are sent in intermediate frames with the last part of the message sent in an end frame.

An intermediate frame is constructed like:

<STX>	FN	text	<ETB>	C1	C2	<CR>	<LF>
-------	----	------	-------	----	----	------	------

An end frame is constructed like:

<STX>	FN	text	<ETX>	C1	C2	<CR>	<LF>
-------	----	------	-------	----	----	------	------

- <STX> = Start of Text transmission control character (ASCII 2)
- FN = Single Digit Frame number ASCII '0' to '7', starting with '1'
- text = Body of message, refer to following chapters
- <ETB> = End of Transmission Block transmission control character (ASCII 23)
- <ETX> = End of Text transmission control character (ASCII 3)
- C1 = Most significant character of checksum 0 to 9 and A to F
- C2 = Least significant character of checksum 0 to 9 and A to F
- <CR> = Carriage Return ASCII character (ASCII 13)
- <LF> = Line Feed ASCII character (ASCII 10)

CLSI LIS1-A standard (see [1], paragraph 6.3.1 *Frame number*), the frame number permits the receiver to distinguish between new and retransmitted frames. The frame number is an ASCII digit ranging from 0 to 7. The frame number begins at 1 with the first frame that is transmitted *within the transfer phase*. The frame

number is incremented by one for every new frame transmitted *within the transfer phase*. After 7 frames have been transmitted, the frame number is set to 0 and continues in this fashion.

The checksum is calculated by adding the binary values of the characters in the frame and keeping the least significant 8 bits of the result. Calculation of the checksum starts with the frame number and ends with the <ETB> or <ETX> character. The calculation of the checksum does not include:

- the starting <STX> character.
- the checksum characters C1 and C2
- the terminating <CR> and <LF> characters

4. Messages

4.1 General

This section describes the format of messages between the PCA and the LIMS. These messages are formatted according to the CLSI LIS2-A standard (see [2]).

Warning:

Messages can both be sent spontaneously and as a response to a query for information.

Each message consists of one or more records (see section 5), each record is terminated with a carriage return character (ASCII 13). The following example shows the flow of control of a message from a sender to a receiver. The characters below ASCII 32 are written in the notation: S_{Tx} (ASCII 2). An ASCII code chart is provided in Annex X1 of the CLSI LIS1-A standard (see [1]), for more examples, refer to Appendix A.

```

Sender:      ENQ
Receiver:    ACK
Sender:      STx1H|\^&|||PrOM^4.2.0|||2.1|||P|LIS2-A|20100421141450CRQ|1|^12936-
            A|ALL|||OCRL|1|FCREETx23CLF
Receiver:    ACK
Sender:      EOT
    
```

A LIMS may transmit more fields and records than are specified in this document. The PCA shall ignore those fields and records as long as the required fields by the PCA are received.

If the PCA receives a message that does not contain the required fields, the PCA shall ignore the complete message.

The PCA verifies the correct sequential numbering of records from the LIMS. The record sequence number and record type fields are therefore required fields by the PCA in the message.

The frame number, as defined in Chapter 3 of the CLSI LIS1-A standard (see [1]), must be sequentially numbered and this is checked by the PCA. Incorrect frame numbering will lead to rejection of the frame.

4.1.1 UTF-8 instead of ISO/IEC 8859-1 (aka Latin-1)

The CLSI LIS2-A standard (see [2], paragraph 6.1 Character set) prescribes that *all data shall be represented as eight-bit, single-byte, coded graphic character values as defined in ISO 8859-1:1987. The eight-bit values, within the range from 0 to 127 of ISO 8859-1:1987 correspond to the ASCII standard character set. Values from 0 to 31 are disallowed with the exception of 7 (BEL), 9 (Horizontal tab), 11 (vertical tab) and 13 (CR), where 13 is reserved as a record terminator. Values from 32 to 126 are allowed as are values from 128 to 254, but the PCA will **not** support the values 128 to 160.*

The PCA supports Unicode for user defined information in non-English languages. This Unicode can be transmitted via UTF-8 encoding. This technique does not violate the character-oriented protocol and associated character restrictions described by the data link layer of the CLSI LIS1-A standard except for the control characters 128 to 160.

UTF-8 is able to represent any character in the Unicode standard, yet is backwards compatible with ASCII (for the first 128 characters a single-byte encoding is used, which identical to the seven-bit ASCII code charts mentioned in the LIS1-A specification reference X1.1).

Disallowed characters (as per LIS1-A 8.6 *Restricted Message Characters*)

The LIS1-A data link protocol is designed for sending character-based message text. Restrictions are placed on which characters may appear in the message text. The restrictions make it simpler for

senders and receivers to recognize replies and frame delimiters. Additional characters are restricted to avoid interfering with software controls for devices such as multiplexers.

An <LF> character is not permitted to appear in the message text; it can appear only as the last character of a frame. None of the ten transmission control characters, the <LF> format effector control, or four device control characters may appear in the message text.

The restricted characters are: <SOH>, <STX>, <ETX>, <EOT>, <ENQ>, <ACK>, <DLE>, <NAK>, <SYN>, <ETB>, <LF>, <DC1>, <DC2>, <DC3> and <DC4>.

Characters that can be supported

The following groups the types of characters supported by Unicode into categories and will be used to describe field contents, to know what to expect/send and to have some form of restriction.

- *Alpha* (Alphabetic): Any alphabetic based system of characters (such as English, French, Spanish and Latin)
- *Logo* (Logographic): Any grapheme based system of characters (such as Chinese, Japanese and Korean)
- *Numeric*: Arabic number characters (0-9)
- *Symbol*: Symbol characters (such as %, # and @)

There are also some fields which limit the characters that will be supported to ASCII (first 128 ASCII characters described in the seven-bits ASCII code charts in the LIS1-A specification). The following will be used to describe these fields.

- *ASCII*: English alphabet set of characters (A-Z), Arabic number characters (0-9), and Symbol characters (such as =, ? and @) which can be represented in ASCII.

It is possible for a field to support a combination of the above types of characters; the applicable descriptors for field content are added to the record tables.

- *Alpha* : only Alphabetic
- *Alpha, Logo* : both Alphabetic and Logographic
- etc

4.2 Record format

The message format texts follow CLSI LIS2-A standard (see [2]).

All records are formatted with delimiters. For ELITech clinical systems, the following delimiters are used (these are send in the header record which each message contains):

Field delimiter: Vertical bar (|) Latin-1 (124)
 Repeat delimiter: Backslash (\) Latin-1 (96)
 Component delimiter: Caret (^) Latin-1 (94)
 Escape delimiter: ampersand (&) Latin-1 (38)

The following escape sequences are recognized and transmitted by the PCA:

&F& imbedded field delimiter character
 &S& imbedded component delimiter character
 &R& imbedded repeat field delimiter character
 &E& imbedded escape delimiter character

In cases where fields are limited to a maximum (e.g., sample_ID, patient name), this maximum applies to the number of characters without replacing imbedded field delimiters by their escape sequence.

Example:

Sample ID "15\&a" is transmitted as "15&R&a"; but the number of characters for maximum limit is 4. (For the maximum number of characters to be used in a frame the escape sequence &R& counts as 3 characters not 1.)

All the text fields are case-sensitive. This means that the PCA performs all text comparisons sensitive to the case of the text field. For instance a request containing a character 's' in the priority field, will not be interpreted as 'S' (STAT) and a test request 'Asat' will not match with a test called 'ASAT' in the PCA.

5. Messages between LIMS and PCA

This section describes the messages that are sent between the LIMS and the PCA. For LIMS implementation, the minimally required fields are marked 'yes', fields marked 'no' are optional. The numbering of the fields is identical to the numbering used in the CLSI LIS2-A standard (see [2]).

Fields marked not required do not necessarily have to be send after the last required field form a message is send. As an example; in the message P (patient information record) field 8.1.9 (patient sex) is the last required field to be send, all fields after 8.1.9 can be left out of the message but might be left in.

Field value identifier *<empty>* means the field must be empty, field value identifier *<ignored>* means that the field will not be used whatever the value (if any). Keep in mind that in future usage the fields identified with *<ignored>* or not required might be used, good practice is to implement these already.

5.1 Header message (H)

Every message contains a header record, the header record contains the following fields:

LIS2-A field					
ID	name	PCA to LIMS	LIMS to PCA	Required	Comment
7.1.1	Record Type ID	H	H	yes	
7.1.2	Delimiter Definition	\ ^ &	\ ^ &	yes	Other field delimiters are not supported.
7.1.3	Message Control ID	<i><empty></i>	<i><ignored></i>	no	
7.1.4	Access Password	<i><empty></i>	<i><ignored></i>	no	
7.1.5	Sender Name or ID	device_id^ software_version	<i><ignored></i> or host_id	yes	device_id: text defined in communication parameters of PCA. software_version: text containing current software version of sending application. host_id: text identifying the LIMS.
7.1.6	Sender Street Address	<i><empty></i>	<i><ignored></i>	no	
7.1.7	Reserved Field	<i><empty></i>	<i><ignored></i>	no	
7.1.8	Sender Telephone Number	<i><empty></i>	<i><ignored></i>	no	
7.1.9	Characteristics of Sender	2 . 2	<i><ignored></i>	yes	Currently represents the version number of the LIMS LIS2-A protocol specification (this document).
7.1.10	Receiver ID	host_id	<i><ignored></i> or device_id	yes	device_id: text defined in communication parameters of PCA. host_id: text identifying the LIMS. (see note below)
7.1.11	Comment or Special Instructions	<i><empty></i>	<i><ignored></i>	no	
7.1.12	Processing ID	P	<i><ignored></i>	yes	
7.1.13	Version No.	LIS2-A	<i><ignored></i>	yes	
7.1.14	Date and Time of	current date/time	<i><ignored></i>	yes	YYYYMMDDHHMMSS

	Message				
--	---------	--	--	--	--

device_id

The field device_id is used to identify the ELITech clinical system and can be used as an extra security to ensure that the tests are requested\executed on the correct ELITech clinical system. It can be up to 20 characters long.

7.1.2 Delimiter Definition

This field specifies the to be used delimiters in decoding the message, between the Record Type ID and the delimiter definition no delimiter is placed because these are not known before this point.

7.1.5 Sender Name or ID

If a host_id is defined in the PCA, and a host_id is present in the received message, then the message will be ignored if the host_id in the received message is not equal to the host_id in the PCA.

Note: that if the host_id in the message or the host_id in the PCA is empty, the just defined rule does not apply. A field may contain more components, these will be ignored.

7.1.10 Received ID

If a device_id is defined in the PCA, and a device_id is present in the received message, then the message will be ignored if the device_id in the received message is not equal to the device_id in the PCA.

Note:that if the device_id in the message or the device_id in the PCA is empty, the just defined rule does not apply. A field may contain more components, these will be ignored.

Example (header from PCA):

H|\^&|||ProM^4.2.0|||2.2|SIM-LIMS||P|LIS2-A|20100421141450

5.2 Patient identifying record (P)

The patient identifying record contains the patient information. Every request from the LIMS starts with a patient identifying record, followed by one or more test order records (see section 5.3). Results from the PCA start with a patient identifying record, followed by one or more test order records, where each order record is followed by one or more result records (see section 5.4). Full examples of messages are provided in Appendix A.

LIS2-A field					
ID	name	PCA to LIMS	LIMS to PCA	Required	Comment
8.1.1	Record Type ID	P	P	yes	
8.1.2	Sequence Number	number	number	yes	Sequential number starting with 1 and continuing until the last patient in the message
8.1.3	Practice Assigned Patient ID	<empty>	<ignored>	no	
8.1.4	Laboratory Assigned Patient ID	<empty>	<ignored>	no	
8.1.5	Patient ID No. 3	medical_record_nr	medical_record_nr	no	maximum 20 characters
8.1.6	Patient Name	patient_name	patient_name	yes	maximum 20 characters see below
8.1.7	Mother's Maiden Name	<empty>	<ignored>	no	
8.1.8	Birthdate	date_of_birth	date_of_birth	yes	YYYYMMDD
8.1.9	Patient Sex	M F	<empty> M	yes	The contents of this field define the applicable

		U	F U		reference limits: M: male F: female U: pediatric Default: M
8.1.10	Patient Race – Ethnic Origin	<ignored>	<ignored>	no	
8.1.11	Patient Address	<ignored>	<ignored>	no	
8.1.12	Reserved Field	<ignored>	<ignored>	no	
8.1.13	Patient Telephone Number	<ignored>	<ignored>	no	
8.1.14	Attending Physician ID	<ignored>	<ignored>	no	
8.1.15	Special Field 1	date_collected	date_collected	no	Note: for backwards compatibility. Information should be retrieved from or sent in field 9.4.8!
8.1.16	Special Field 2	<ignored>	<ignored>	no	
8.1.17	Patient Height	<ignored>	<ignored>	no	
8.1.18	Patient Weight	<ignored>	<ignored>	no	
8.1.19	Patient's Known or Suspected Diagnosis	<ignored>	<ignored>	no	
8.1.20	Patient Active Medications	<ignored>	<ignored>	no	
8.1.21	Patient's Diet	<ignored>	<ignored>	no	
8.1.22	Practice Field No. 1	<ignored>	<ignored>	no	
8.1.23	Practice Field No. 2	<ignored>	<ignored>	no	
8.1.24	Admission and Discharge Dates	<ignored>	<ignored>	no	
8.1.25	Admission Status	<ignored>	<ignored>	no	
8.1.26	Location	<ignored>	<ignored>	no	
8.1.27	Nature of Alternative Diagnostic Code and Classifiers	<ignored>	<ignored>	no	
8.1.28	Alternative Diagnostic Code and Classifiers	<ignored>	<ignored>	no	
8.1.29	Patient Religion	<ignored>	<ignored>	no	
8.1.30	Marital Status	<ignored>	<ignored>	no	
8.1.31	Isolation Status	<ignored>	<ignored>	no	
8.1.32	Language	<ignored>	<ignored>	no	
8.1.33	Hospital Service	<ignored>	<ignored>	no	
8.1.34	Hospital Institution	<ignored>	<ignored>	no	
8.1.35	Dosage Category	<ignored>	<ignored>	no	

Behaviour of the ELITech clinical system upon receiving patient information; the PCA will keep any information that was entered locally and ignore any details from the LIMS (if they are different). If there are **no** local details, the PCA will use the LIMS request to fill-in the sample details.

8.1.6 Patient name

The patient name field is to be filled with one component and one component only. This component shall be no longer than 20 characters and may include special characters.

8.1.9 Patient Sex

In non-clinical laboratories, meaning that the samples measured are other than human ('male', 'female' or 'pediatric'). For instance it could be a plant or animal, meaning that these would also have different names for the reference limits. Between the PCA and LIMS always three types of reference limits are communicated and these are fixed to **M**, **F** and **U** respectively.

Warning: If you use the ELITech clinical system for non human samples make sure you are aware of the correct translation between the PCA and LIMS for the patient sex field.

Example (patient information record from PCA):

P|1||||Henry Johnson||19650714|M

5.3 Test order record (O)

The order record contains information on the test to be performed for the preceding patient record (see section 5.2).

LIS2-A field					
ID	name	PCA to LIMS	LIMS to PCA	Required	Comment
9.4.1	Record Type ID	O	O	yes	
9.4.2	Sequence Number	number	<ignored>	yes	Sequential number starting with 1 and continuing until the last order for the patient
9.4.3	Specimen ID	<empty> sample ID	<empty> sample ID	yes	Maximum 11 characters Empty for calibrator, control and blanks.
9.4.4	Instrument Specimen ID	<empty> Calibrator. name Control name BLANK	<empty> Calibrator. name Control name BLANK	yes	For PCA 'samples' like calibrators, controls, and blank measurements, the specimen ID field (9.4.3) will be empty to avoid conflicts. This field identifies calibrator by name; control by name or blank. See notes below
9.4.5	Universal Test ID	<empty>	^^abbr_test or <empty>	yes	Abbreviated test name max 4 characters long. LIMS may send multiple requests in single order record by use of repeat delimiter. Calculated tests can also be requested on abbreviated test name. See notes below
9.4.6	Priority	S A R	<empty> S A R	yes	S (Stat) A (ASAP / Pediatric) R (Normal) All other will be treated as R . Default: R
9.4.7	Requested/Ordered Date and Time	<empty>	<ignored>	no	
9.4.8	Specimen Collection Date	<empty> Collection_dateti	<empty> Collection_date	no	YYYYMMDDHHMMSS, parser can currently ignore

	and Time	me	time		time as this is not set / used in the PCA.
9.4.9	Collection End Time	<empty>	<ignored>	no	
9.4.10	Collection Volume	<empty>	<ignored>	no	
9.4.11	Collector ID	<empty> Collector_ID	<empty> Collector_ID	no	
9.4.12	Action Code	<empty>	<empty> N A C	yes	Empty and other will be treated as N See notes below
9.4.13	Danger Code	<empty>	<ignored>	no	
9.4.14	Relevant Clinical Information	<empty>	<ignored>	no	
9.4.15	Date/Time Specimen Received	<empty>	<ignored>	no	
9.4.16	Specimen Descriptor	sample type BLANK CALIBRATOR	<empty> sample type	yes	Sending of reagent blank and calibrator results is not supported in current versions of the Touch Pro software, but will be added in future. Control samples have a sample type, just like patient samples. Also see notes below
9.4.17	Ordering Physician	<empty> Name	<empty> name	yes	Max 20 characters. In case this field contains multiple components, only the first component is used.
9.4.18	Physician's Telephone Number	<empty>	<ignored>	no	
9.4.19	User Field No. 1	<empty>	<ignored>	no	
9.4.20	User Field No. 2	<empty>	<ignored>	no	
9.4.21	Laboratory Field No. 1	<empty>	<ignored>	no	
9.4.22	Laboratory Field No. 2	<empty>	<ignored>	no	
9.4.23	Date/Time Results Reported or Last Modified	<empty>	<ignored>	no	
9.4.24	Instrument Charge to Computer System	<empty>	<ignored>	no	
9.4.25	Instrument Section ID	<empty>	<ignored>	no	
9.4.26	Report Types	F X I	<empty> O Q Y Z	yes	See notes below

9.4.27	Reserved Field	<ignored>	<ignored>	no	
9.4.28	Location or Ward of Specimen Collection	<ignored>	<ignored>	no	
9.4.29	Nosocomial Infection Flag	<ignored>	<ignored>	no	
9.4.30	Specimen Service	<ignored>	<ignored>	no	
9.4.31	Specimen Institution	<ignored>	<ignored>	no	

9.4.4 Instrument Specimen ID

When a calibrator, control or blank is requested from the LIMS, fields with sample information will be ignored.

9.4.5 Universal Test ID

Can be empty when sent from LIMS to PCA in case of a host query result being that the requested sample ID is not known or in case of an Action code cancel request is sent.

To request ISE tests, the **abbreviate name is required**. The PCA reports ISE results as a single Test order record and separate result records for each ISE result

Example (ISE request from LIMS):

```
O|1|12935-C||^K+|R||||N|||Serum|Dr. CarnegyCR
O|2|12935-C||^Na+|R||||A|||Serum|Dr. CarnegyCR
O|3|12935-C||^Cl-|R||||A|||Serum|Dr. CarnegyCR
```

Example (ISE result from PCA):

```
O|1|12935-C||R|||||Dr. Carnegy|||FCR
R|1|^K+^Potassium|4.2|mmol/l|||F|||20100428123012CR
R|2|^Na+^Sodium|139|mmol/l|||F|||20100428123012CR
R|3|^Cl-^Chloride|111|mmol/l|||F|||20100428123012CR
```

To request calculated tests, the value for the abbr_test must be the abbreviated name known at the PCA. The PCA reports calculated test results as a single Test order record and separate results records for each Calculated result, containing '^Calc^[item]', etc. in the result record.

Example (Calculated test result from PCA):

```
O|1|12935-C||R|||||Dr. Carnegy|||FCR
R|1|^LCL^LDL Cholesterol|122|mmol/l|||F|||20100503123021CR
```

Furthermore;

if an order is made for the calculated test only, the result record will hold the result of the calculated test and all underlying tests. The underlying test results may be suppressed in later versions of the Touch Pro software. If an order is made with the calculated test and all underlying tests (meaning all tests belonging to the calculated test) the result record will hold the calculated result and all underlying tests.

If only the underlying tests (meaning all tests belonging to a calculated test) are ordered the result record will hold all the underlying tests (meaning no calculated tests).

9.4.12 Action code

N New request: If a request with the same Sample ID already exists, that request will be overwritten. If the sample is in process or ready, and still loaded, a new request is made and the name, birth date, etc. must be equal to the sample that is in process. Otherwise it would mean that multiple patient's samples are on the instrument with exact same sample IDs (see also 9.1.26 Report Type). New request should also be used to delete requests. Because the New request always sends all of the assay for a sample to be executed, remove one assay from that sample request and send them all again. The sample request will be overridden and the specific assay is removed.

- A** Additional requests: The contents of the order record will be appended to an existing sample request on the PCA. This means that the set of test requests is extended with the test requests in the order record. If a request does not exist or was already started a new request is created.
- C** Cancel: will remove the entire sample request from the request buffer. The contents of field (9.4.5) will be ignored if **C** is specified. (Keep in mind that this will only work up until the moment a request is in process.)

If no action code is provided, the order will be treated as a *New Request* in the PCA.

9.4.16 Specimen Descriptor

This field contains in full text the sample type (or is empty), for instance if the sample type is urine, this field contains **Urine**. The text is compared case sensitive to the sample types defined on the PCA. If the sample type is not known or not defined, it will default to **Serum**.

When transmitted by the PCA, this field can contain any of the defined sample types.

Example (Order record with sample type from PCA):

```
O|1|12935-C|||R|||||||Urine|Dr. Carnegy|||||||FCR
```

9.1.26 Report Type (received from PCA):

- F** Final: Indicates a result is following and that it is the last result for the sample.
- I** Intermediate: Indicates a result is following and that the sample still has active measurements.
- X** Used to indicate a request to measure is not accepted by the PCA. The request may be rejected for the following reasons:
- Sample_ID already exists, but demographic information doesn't match, i.e., Patient name and birth date are not equal to an existing request.
 - No space for request, i.e., the request buffer is full.
- No results will be transmitted for the request.

9.1.26 Report Type (send to PCA):

The field will be ignored if action code (field 12) is C.

- O, Q, <empty>** The requests will replace an existing sample request.
- Y, Z** A LIMS can use these values in response to a 'host query' to indicate that a Sample ID is not known at the LIMS (also the message terminator record Field 13.1.3. is used for this, see 5.7). The operator will be informed of this by the PCA
- <others>** The record will be ignored.

Limitations:

The sample ID in the order record is used to identify the sample request on the PCA, cancelling the order means that the sample request – including patient information – is removed from the request buffer in the PCA. (Keep in mind that this will only work up until the moment a request is in process.)

If the analyser is active when the request is received and the sample is already loaded on the analyser, the requested test is **not** automatically started. (In the rare case where the lid is opened and closed and PSID is installed the sample could be started after a scan.)

If a request (identified with a sample ID) has a name, birth date or other demographic information that conflicts with a request in the PCA, the request will be rejected by the PCA.

Example (Order record from PCA):

```
O|1|12934-A|||^CHOL|R|||||||Dr. Carnegy|||||||FCR
```

5.4 Result record (R)

For every order record the PCA will report one or more result records.

LIS2-A field

ID	name	PCA to LIMS	LIMS to PCA	Required	Comment
10.1.1	Record Type ID	R	This record is always ignored when received from LIMS	yes	
10.1.2	Sequence Number	number		yes	Sequential number starting with 1 and continuing until the last result for the patient.
10.1.3	Universal Test ID	^^abbr_test ^full_test		yes	Abbreviated test name up to 4 characters long; See notes below
10.1.4	Data or Measurement Value	Result^Cut-off evaluation		yes	See notes below
10.1.5	Units	units		yes	Field representing the units into which the results is calculated. Keep in mind that all transmitted data is in the ISO/IEC 8859-1 aka Latin-1 character coding standard.
10.1.6	Reference Ranges	cutoff_reference^ low_reference_limit^ high_reference_limit^ expected_separation		yes	All components are floating point values. cutoff_reference contains the relevant reference value for positive/negative reporting. low_reference_limit and high_reference_limit define the applicable reference range (male/female/pediatric) for the sample. expected-separation defines the applicable separation value for controls for qualitative tests.
10.1.7	Result Abnormal Flags	normalcy_flags^ instrument_flags		yes	normalcy flags: L : Below low reference limit LL: Below low panic limit H : Above high reference limit HH: Above high panic limit < : Below absolute limit > : Above absolute limit Instrument flags: See Appendix B for an overview of possible values
10.1.8	Nature of Abnormality Testing	<empty>		yes	
10.1.9	Result Status	F I		yes	F indicates a final result For future reference: I indicates a result is pending (Field 10.1.4 will contain BUSY or -----). Current software never sends intermediate results.

		x			X indicates the order cannot be done, it will be transmitted immediately upon request to perform tests that are not known at the PCA, the result will show UNKNOWN .
10.1.10	Date of Change	<empty>		no	
10.1.11	Operator Identification	Operator_ID		no	Operator logon name that started the test on the instrument (max length 50 characters).
10.1.12	Date/Time Test Started	<empty>		no	
10.1.13	Date/Time Test Completed	meas_datetime		no	YYYYMMDDHHMMSS Good practice is sending this record always on a results status 'F'.
10.1.14	Instrument Identification	<empty>		no	

10.1.3 Universal Test ID

For ISE results, the field is formatted as follows:

```
^^^K+^Potassium
^^^Na+^Sodium
^^^Cl-^Chloride
^^^CO2^Carbon Dioxide
```

^^^abbreviate name^<full name>

Refer to section 5.3 field 9.4.5 for an example.

For Calculated results, the field is formatted as follows:

```
^^^LCL^LDL Cholesterol
```

^^^<calculated test abbreviation name>^<calculated test name>

Refer to section 5.3 field 9.4.5 for an example.

10.1.4 Data or Measurement value

The field can contain the following:

Floating point number^Cut-off evaluation	The measured value in human readable format, the decimal separator is a period (.), and for qualitative or semi-quantitative the indicator whether the result is POSITIVE or NEGATIVE. Second component field is not sent for quantitative tests. Field 10.1.7 will contain the applicable instrument_flags (+ or -)
REJECT	The test was rejected (either automatically by the PCA, or manually by the operator)
BUSY	The test is being measured at the analyser (note: not yet implemented).
-----	The test is waiting for operator input. (note: not yet implemented)
> n.nnnn	Measurement above assay range. Where n.nnn is a floating point value representing the highest concentration level defined in the test parameters.
< n.nnnn	Measurement below analytical sensitivity. Where n.nnnn is a floating point value representing the lowest concentration level defined in the test parameters.
UNKNOWN	The test request was unknown at the PCA. Results records reporting UNKNOWN to the LIMS will not show in the result list of the PCA.

10.1.7 Result abnormal flags

The component normalcy_flags will specify '<' or '>' if the value is outside a range other than a reference range. The component instrument_flags will provide more detail of the range that was exceeded. The following table lists the instrument flags and the normalcy flags that will be set accordingly:

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Instrument flags		Normalcy flag
n	Reference low error	L
y	Reference panic low error	LL
N	Reference high error	H
Y	Reference panic high error	HH
m	Absorbance low limit error	<
U	Measurement counter underrange	
u	Reference counter underrange	
k	Measurement below low calibration limit	
x	Measurement below analytical sensitivity	
M	Absorbance high limit violation	>
O	Measurement counter overrange	
o	Reference counter overrange	
K	Measurement above high calibration limit	
X	Measurement above assay range	

General

By default the PCA sends the results to the LIMS as soon as they become available. This means that the PCA will send multiple result messages for a single sample (1 for each test result). It also means that if there is a test that requires operator attention (the test has an *INFO*), then this test will not hold back sending of results. The field *Report Type* (9.1.26) is used to indicate if the sample has still has active tests that are being measured, or if it is the last test for the sample.

Alternatively, the option *Collate Results* may be set in the PCA, in which case the PCA will not send the test results for a sample, until all tests for that sample are completed. It will then send all results in a single message.

In case the test measurement is performed with a repeat value > 1 (the test must have been requested at the PCA console, since the LIMS request does not support specifying a repeat value), the average result is reported to the LIMS. In future, each individual measurement may be reported to the LIMS. The LIMS can then identify these results as different measurement by inspecting the value of the field *measurement date/time* (10.1.13).

Example (result record from PCA):

```
R|1|^C300^+Cocaine 300|0.276|dAbs/m|0.301|^-
|F|||20060120153902|L^Z2820^20060701^C_R
```

5.5 Comment record (C)

The format of the comment record depends on the context in which it is received or transmitted.

5.5.1 Comment for Patient information record (P)

Following a patient information record there can be a comment record containing a specific comment for the sample.

LIS2-A field					
ID	name	PCA to LIMS	LIMS to PCA	Required	Comment
11.1.1	Record Type ID	c	c	yes	
11.1.2	Sequence Number	1	<ignored>	yes	Always 1, because only one comment record is allowed for patient information record. See notes below

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11.1.3	Comment Source	I	<ignored>	yes	
11.1.4	Comment Text	text	text	yes	see below
11.1.5	Comment Type	G	<ignored>	yes	

According to LIS2-a standard (see FIG. 1 Logical Structure of a Message) it is only possible to add one comment record for a patient information record.

If the PCA receives a comment record following the patient information record, the text from the comment record will be placed in the sample comment record.

11.1.4 Comment text

The PCA will store only the first 100 characters of the received comment text.

Example (comment order record from PCA):

C|1|I|Patient is dying |G

5.5.2 Comments for Result records (R)

If 'extended results' is enabled in communication settings, the PCA will add comment records following the result record. These comment records contain raw result information for the transmitted result.

LIS2-A field					
ID	name	PCA to LIMS	LIMS to PCA	Required	Comment
11.1.1	Record Type ID	C	The comment record following the result record is always ignored when received from the LIMS	yes	
11.1.2	Sequence Number	1		yes	Sequential number starting with 1 and continuing until the last comment on this level.
11.1.3	Comment Source	I		yes	
11.1.4	Comment Text	code^text		yes	see below
11.1.5	Comment Type	G		yes	

C|<sequence number>|I|M01^<abs_result>^<abs_units>|G
 <abs_result> Calculated absorbance value based on raw measurements.
 <abs_units> Relevant absorbance units (Abs, dAbs, dAbs/m)

C|<sequence number>|I|M02^<seconds>^<raw_result>^<used>|G
 <seconds> Time at which measurement was performed. This is a floating point value.
 <raw_result> floating point value specifying the raw absorbance measurement by the analyser
 <used> 0 or 1: 1 indicates the point is used by the PCA to calculate the concentration; 0 indicates the measurement was not used.

C|<sequence number>|I|M03^<reagent_batch_number>^<reagent_expiry_date>|G
 <reagent_batch_number> The reagent batch number only of the first reagent of the test (R1). The reagent batch number shall not be send for calculated tests.
 <reagent_expiry_date> The reagent expiry date only of the first reagent of the test (R1). The reagent expire date shall not be send for calculated tests (YYYYMMDD).

11.1.4 Comment text

Raw measurement results (M01 and M02)

The field *Comment Text* is repeated for every raw measurement point.

In case a 'test repeat' is set in the test parameters, the raw measurements are sent for each repeat. With a test repeat of 3, this means that 6 comment records will be appended for each result.

Raw results from the ISE are reported in the format as they are received from the ISE unit. Currently the ISE units reports only final results. Only an M01 field will be transmitted.

Example (comment result records from PCA):

C|1|I|M01^0.276^dAbs/m|G

C|2|I|M02^24.5^0.2313^0\M02^51^0.2320^0\M02^77.5^0.7123^0\[additional values skipped for brevity]|G

Reagent information (M03)

The field *Comment Text* is repeated in case of more than one batch for the first reagent of the test (R1) is used.

Example (comment result records from PCA):

C|1|I|M03^101453^20120131\M03^101717^20110630|G

5.6 Request information record (Q)

The request information record is used by either the LIMS or the PCA to remotely request information from the other party.

LIS2-A field					
ID	name	PCA to LIMS	LIMS to PCA	Required	Comment
12.1.1	Record Type ID	Q	Q	yes	
12.1.2	Sequence Number	1	1	yes	Sequential number starting with 1 and continuing until the last request for this message
12.1.3	Starting Range ID Number	^sample_ID	^sample_ID or ALL	yes	In case ALL is received by the PCA; the PCA will (re)send the last 100 results in the result buffer.
12.1.4	Ending Range ID Number	<empty>	<ignored>	no	Only a single sample ID can be requested in each query record.
12.1.5	Universal Test ID	ALL	<ignored>	no	By default ALL results for a sample are transmitted.
12.1.6	Nature of Request Time Limits	<empty>	<ignored>	no	
12.1.7	Beginning Request Results Date and Time	<empty>	<ignored>	no	
12.1.8	Ending Request Results Date and Time	<empty>	<ignored>	no	
12.1.9	Requesting Physician Name	<empty>	<ignored>	no	
12.1.10	Requesting Physician Telephone Number	<empty>	<ignored>	no	

12.1.1.1	User Field No. 1	<empty>	<ignored>	no	
12.1.1.2	User Field No. 2	<empty>	<ignored>	no	
12.1.1.3	Request Information Status Codes	O	<ignored>	no	

Whenever the PCA encounters a new sample ID for which it cannot find a request in the request buffer, it sends a 'request information record' to the LIMS, requesting to send test orders and demographics. The PCA will never automatically cancel an outstanding sample request (it will be kept in the request buffer).

When the PCA receives a request it will always respond by sending all results for the requested sample ID as they are at that time. These results may be Final (F) or in PCA pending (I).

5.7 Message Terminator Record (L)

Every message must be terminated with a Message Terminator record. Since every frame can only contain one message, this also terminates the frame (see section 3.2).

LIS2-A field					
ID	name	PCA to LIMS	LIMS to PCA	Required	Comment
13.1.1	Record Type ID	L	L	yes	
13.1.2	Sequence Number	1	1 <ignored>	yes	Always 1
13.1.3	Termination Code	F	F Q I	yes	

When the Termination Code (13.1.3) received from the LIMS is Q or I then this means that the LIMS has no information on the sample ID in the last request, the operator will be informed of this by the PCA.

5.8 Manufacturer Information Record (M)

This record is used for custom information exchange not covered in the CLSI LIS2A standard (see [2]).

For ELITech clinical systems this custom information is:

- Sending cuvette blank information to the LIMS
- Requesting full version information of all analyser parts
- Downloading error history information from PCA
- Reagent installation information and statistics
- Test installation information

These information records are always level 1 records

LIS2-A field					
ID	name	PCA to LIMS	LIMS to PCA	Required	Comment
15.1.1	Record Type ID	M	M	yes	
15.1.2	Sequence Number	1	1 <ignored>	yes	Only 1 manufacturer information record is allowed for each message.

5.8.1 Cuvette Blank information (B)

The cuvette blank information is sent by the PCA on request of the operator using the PCA (in the blank rotor menu). The blank results of the selected filter and analyser half are sent to the LIMS. The PCA will send the cuvette blank information in one message and not multiple.

If the LIMS requests blank information by sending an M|1|B, then all blank information, for all cuvettes and all wavelengths is transmitted to the LIMS.

LIS2-A field					
ID	name	PCA to LIMS	LIMS to PCA	Required	Comment
15.w.3	Manufacturer Record Type ID	B	B	yes	
15.w.4	Analyser part	<empty> L R	<ignored>	no	One analyser part: <empty> Multiple analyser parts: Left or Right analyser part.
15.w.5	Wavelength	<wavelength>	<ignored>	yes	Wavelength in nanometres.
15.w.6	Cuvette blank value	<value>	<ignored>	yes	Absorbance value for each cuvette, the value is repeated 48 times.
15.w.7	Cuvette average	<value>	<ignored>	yes	
15.w.8	Cuvette Standard Deviation	<value>	<ignored>	yes	
15.w.9	Lamp average	<value>	<ignored>	yes	
15.w.10	Lamp Standard Deviation	<value>	<ignored>	yes	

Example (manufacturer information record with cuvette blank information from PCA):

```
H|\^&|||SELXL^2.1.0|||2.1|||P|LIS2-A|20051221153902CR
M|1|B|L|340|0.1029\0.1030\0.1041\0.1032<44 values skipped for
brevity>|0.1033|0.0012|0.0291|0.0009CR
L|1|FCR
```

5.8.2 Error History (H)

Warning: this record is primarily intended for use during production and service of the analyser and is subject to change.

The error history is only transmitted to the LIMS after the LIMS has requested the error history. The LIMS performs an error history request by sending an M|1|H record.

The following fields are appended to the Manufacturer Information Record:

LIS2-A field					
ID	name	PCA to LIMS	LIMS to PCA	Required	Comment
15.x.3	Manufacturer Record Type ID	H	H	yes	
15.x.4	Error History record	error_datetime^ error_message^ operator_action	<ignored>	yes	Field is repeated for each item in error history
15.x.5	Action code	<empty>	<empty> C	no	C: Clear error history

Component	Content	Length	Comment
error_datetime	YYYYMMDDHHMMSS	14	date and time of error
error_message	text	5	Error code (e.g., E102)

operator_action	digit	1	A code identifying the action of the operator: 0 - None 1 - Check Again 2 - Acknowledge 3 - Reset System 4 - Specific Reset 5 - Remain Inactive 6 - Halt 7 - Analyser side reset
-----------------	-------	---	--

Example (manufacturer information record with error history from PCA):

```
H|\^&|||SELXL^2.1.0|||2.1|||P|LIS2-A|20051221153902CR
M|1|H|20050827151007^E12^0\20050827151210^E17^0\20051009101247^E13^6CR
L|1|FCR
```

5.8.3 Test installation (T)

Test installation information is sent automatically from the PCA to the LIMS whenever the installed test configuration in the PCA changes (mainly driven by the reagent that are installed for an test). Alternatively the LIMS can request test installation information by sending a M|1|T record.

The following fields are appended to the Manufacturer Information Record:

LIS2-A field					
ID	name	PCA to LIMS	LIMS to PCA	Required	Comment
15.u.3	Manufacturer Record Type ID	T	T	yes	
15.u.4	Test installation information	abbr_test^ full_test^ OR <empty>	<ignored>	yes	Field is repeated for each installed test <empty> in case of no test(s) installed.

Component	Content	Max Length	Comment
abbr_test	text	4	Abbreviated test name
full_test	text	15	Full test name

The ISE test will be reported as any other test; once all the liquids needed are installed. The amount of liquid in a reagent vial or if valid calibration and/or control data is available will not be taken into account when sending the test installation information. All tests that have completely been installed will be reported.

Besides ISE and photometric tests also calculated tests will be send by the PCA to the LIMS.

Example (manufacturer information record with test installation from PCA):

The following message indicates 2 tests (ISE, calculated and/or photometric) are installed in the PCA; "+Cocaine 150" and "LSD".

```
H|\^&|||ProM^4.2.0|||2.1|||P|LIS2-A|20060120162301CR
M|1|T|+C15^+Cocaine 150\LSD^LSDCR
L|1|FCR
```

5.8.4 Extended version information (V)

Warning: this record is primarily intended to be used during production and service of the analyser and is subject to change.

The extended version information is only transmitted to the LIMS after the LIMS has performed a request by sending an M|1|V record.

Extended version information can be used to request version information for all parts used in the analyser.

LIS2-A field					
ID	name	PCA to LIMS	LIMS to PCA	Required	Comment
15.z.3	Manufacturer Record Type ID	v	v	yes	
15.z.4	Version information	part^ version	<ignored>	yes	Field is repeated for each analyser part

Component	Content	Length	Comment
part	number	2	Unique number identifying the analyser part: 1 PC release description 2 PC release version 3 PC application version 4 PC application date 5 CCB version (XL-series only) 6 Sample Disc Version 7 Barcode version (XL-series only) 8 (Left) Master version 9 (Left) Master Date 10 (Left) Pipettor version 11 (Left) Measurement unit version 12 (Left) Sample arm version 13 (Left) Reagent arm version (XL and E-series only) 14 (Left) Reagent discs version (XL and E-series only) 15 (Left) Wash arm version 16 Right master version (XL-series only) 17 Right master date (XL-series only) 18 Right Pipettor version (XL-series only) 19 Right Meas. Unit version (XL-series only) 20 Right Sample arm version (XL-series only) 21 Right Reagent arm version (XL-series only) 22 Right Reagent disc version (XL-series only) 23 Right wash arm version (XL-series only)
version	text		Text describing version or date as it was received from the analyser.

Example (manufacturer information record with version information from PCA):

```
H|\^&|||XL400^4.0.0|||2.1|Host||P|LIS2-A|20080103120702^C_R
M|1|V|1^XL Analyser 4.0\2^4.0.0\3^4.0.0\4^20071225\5^XL/VTLV1.3.1\
6^DSC P0.03 \7^8^XL/DRY V1.1.0 (DUAL)\9^14-APR-2004 13:49\
10^DIL V1.10 \11^MDS V1.10 \12^SA V1.10 \13^RA V1.20 \
14^DSC V1.10 \15^WA V1.11 \16^XL/DRY V1.1.0 (DUAL)\
17^14-APR-2004 13:49\18^DIL V1.10 \19^MDS V1.10 \
20^SA V1.10 \21^RA V1.20 \22^DSC V1.10 \23^WA V1.11^C_R
L|1|F^C_R
```

|

Appendix A. Examples of messages

Example of a result message from PCA to LIMS (in non extended format):

```
H|\^&|||ProM^4.2.1|||1.5|||P|LIS2-A|20100617120936^C_R
P|1|||M^C_R
O|1|20100617003||R|||||Normal|||||F^C_R
R|1|^VPA^E2 Valproic Acid|-0.002|dAbs/m|-0.002|<^aVm|F|||20100609110047^C_R
L|1|F^C_R
```

Example of a result message from PCA to LIMS (in extended format):

```
H|\^&|||ProM^4.2.1|||1.5|||P|LIS2-A|20100617120936^C_R
P|1|||M^C_R
O|1|20100617003||R|||||Normal|||||F^C_R
R|1|^VPA^E2 Valproic Acid |-0.002|dAbs/m|-0.002|<^aVm|F|||20100609110047^C_R
C|1|I|M01^-0.002^dAbs/m|G^C_R
C|2|I|M02^-4.00^-0.2797^0\M02^-3.50^-0.2797^0\M02^23.75^-0.0980^1\M02^50.25^-
0.1017^1\M02^76.75^-0.1011^1\M02^103.25^-0.1012^1\M02^129.75^-
0.1013^0\M02^156.25^-0.1007^0\M02^182.75^-0.1015^0\M02^209.25^-
0.1010^0\M02^235.75^-0.1024^0\M02^262.25^-0.1017^0\M02^280.00^-
0.1016^0\M02^280.50^-
0.1016^0\M02^288.75^1.3804^0\M02^307.00^1.3494^0\M02^333.50^1.3358^0\M02^360.00^
1.3291^0\M02^386.50^1.3245^0\M02^413.00^1.3179^0\M02^439.50^1.3106^0\M02^466.00^
1.3033^0\M02^492.50^1.2991^0\M02^519.00^1.2911^0\M02^673.00^-
0.3000^0\M02^673.50^-0.3000^0|G^C_R
L|1|F^C_R
```

Example of a result message for calibration (Valproic Acid assay) from PCA to LIMS (for future reference, not supported yet by Touch Pro software)

```
H|\^&|||ProM^4.2.1|||1.5|||P|LIS2-A|20100617120936^C_R
P|1|||M^C_R
O|1|VPA Calibrator|R|||||F^C_R
R|1|^VPA^E2 Valproic Acid|0.2258|dAbs/m|||F|||20100617081456^C_R
R|2|^VPA^E2 Valproic Acid|0.2524|dAbs/m|||F|||20100617081550^C_R
R|3|^VPA^E2 Valproic Acid|0.2837|dAbs/m|||F|||20100617081644^C_R
R|4|^VPA^E2 Valproic Acid|0.3084|dAbs/m|||F|||20100617081738^C_R
R|5|^VPA^E2 Valproic Acid|0.3631|dAbs/m|||F|||20100617081832^C_R
R|6|^VPA^E2 Valproic Acid|0.3892|dAbs/m|||F|||20100617081926^C_R
L|1|F^C_R
```

Example of a host query with host known sample ID

```
H|\^&|||ProM^4.2.1|||1.5|SIM-LIMS|P|LIS2-A|20100609141144^C_R
Q|1|^2010060901||ALL|||||O^C_R
L|1|F^C_R
```

Example of a host query answer with host known sample ID

```
H|\^&|||SIM-LIMS|||||20100609141147^C_R
P|1|||Patient_1||19980101|M|||||X^C_R
O|1|2010060901||^test|N||||N||||Doc^C_R
O|2|2010060901||^t2|N||||N||||Doc^C_R
L|1|F^C_R
```

Example of a host query with host unknown sample ID

```
H|\^&|||ProM^4.2.1|||1.5|SIM-LIMS|P|LIS2-A|20100609141528^C_R
Q|1|^2010060902||ALL|||||O^C_R
```

L|1|F^{C_R}

Example of a host query answer with host unknown sample ID

H|\^&|||ProM ^4.2.1|||1.5|SIM-LIMS||P|LIS2-A|20100609141528^{C_R}
P|1|||||||^{C_R}
O|1|2010060902|||||||||||||||z^{C_R}
L|1|F^{C_R}

Example of a host query answer with host unknown sample ID (shorter)

H|\^&^{C_R}
L|1|I^{C_R}

Appendix B. 10.1.7; instrument_flags

The following flags will be send from the PCA to the LIMS in the field component instrument_flags of field 10.1.7 result abnormal flags.

Flag Code	Description
a	Reagent absorbance limit violation.
A	Calibrator limit violation.
c	Control limit violation.
C	Control limit violation.
D	Reagent absorbance deviation error.
D	Substrate depletion error.
E	Result near cut-off.
G	General hardware error.
k	Result exceeds calibration limits.
K	Result exceeds calibration limits.
L	Cuvette blank error.
m	Low absorbance limit violation.
M	High absorbance limit violation.
N+	Reference limit violation.
N-	Reference limit violation
o	Reference Over Range Error
O	Over Range Error
r	Rerun (Indicates that the sample has already been rerun)
R-	Insufficient Reagent
R+	Reagent Not Taken
U	Under Range error
u	Reference Under Range error
+	Positive (for qualitative or semi-quantitative tests).
-	Negative (for qualitative or semi-quantitative tests).
w	Westgard rules limit violation.
W	Westgard rules limit violation.
x	Concentration limit error.
X	Concentration limit error.
Y	Reference panic high limit error
y	Reference panic low limit error
Z	Duplicate-difference limit error
*	Non-linearity error (kinetic tests).
#	Insufficient sample.